Practical No. 6: Measures of Central Tendency and Dispersion for Ungrouped Data

Blue = data from question; green = results. Sample variance / S.D. use (n − 1). 'Manual' columns use the formula directly; Excel's built-in functions are shown as a check.

Ex.1 Sample 53, 55, 70, 58, 64, 57, 53, 69, 57, 68, 53: mean, median and mode										
x	53	55	70	58	64	57	53	69	57	68
	53									
Results	**n**	11	**Σx**	657	**Mean**	**59.727**	**Median**	**57.**	**Mode**	**53.**

Expected: 59.727, 57 and 53

Ex.2 Oxidation-induction time (min) of commercial oils: sample range, variance and standard deviation										
Time (min)	87	103	130	160	180	195	132	145	211	105
	145	153	152	138	87	99	93	119	129	
Results	**n**	19	**Range**	**124**	**s^2 (manual)**	**1264.766**	**$s = \sqrt{s^2}$**	**35.564**	**s^2 (VAR)**	1264.766

Expected: range 124; variance 1264.766; S.D. 35.564

Ex.3 Hawaii visitors (thousands): mean, median, range, S.D. and coefficient of variation										
Mainland	108.70	112.25	94.01	144.03	162.44	161.61	76.20	102.11	110.87	79.36
	129.04	95.16	114.16	121.88						
Asia & Pacific	29.89	41.13	40.67	40.41	43.07	24.86	31.61	21.60	27.34	64.57
	32.98	41.31								
Mainland (n=14)	**Mean**	**115.13**	**Median**	**111.560**	**Range**	**86.24**	**S.D.**	**26.82**	**C.V. %**	**23.30**
Asia & Pacific (n=12)	**Mean**	**36.62**	**Median**	**36.695**	**Range**	**42.97**	**S.D.**	**11.40**	**C.V. %**	**31.13**
Comparison	**Greater mean: Mainland; greater S.D.: Mainland; greater relative variation (C.V.): Asia (23.30% vs 31.13%)**									

Expected: Mainland 115.13, 111.56, 86.24, 26.82, 23.30%; Asia 36.62, 36.695, 42.97, 11.40, 31.13%. C.V. = S.D./mean × 100

Ex.4 Days taken by computer manufacturers to resolve problems: mean, median, variance, S.D., best record										
Company	Compaq	Packard Bell	Quantex	Dell	NEC	AST	Acer	Gateway	Digital	IBM
Days	13	27	11	14	14	17	16	21	27	12
Company	Hewlett-Packard	AT&T	Toshiba	Micron						
Days	14	20	37	17						
Results	**n**	14	**Mean**	**18.571**	**Median**	**16.5**	**Variance s^2**	**53.495**	**S.D. s**	**7.314**
Best record	**Fewest days**	11		**Quantex**						

Expected: mean 18.571, median 16.5; variance 53.494, S.D. 7.314; best record: Quantex (11 days)